

BIOL-8 Exam 01

College of the Redwoods, Del Norte

Instructor: Dr. Daniel Friedman

Name: _____

Date: _____

Modules 01-06

Date: Thursday, February 12, 2026

Time: 5:30 PM - 8:40 PM

Total Points: 50

Part A: Multiple Choice (30 points)

Select the best answer for each question. Each question is worth 1 point.

Module 01: Exploring Life Science

1. Which feature is NOT a universal characteristic of living things?

- A) Ability to evolve over time
- B) Reproduction and growth
- C) Dependence on photosynthesis for energy
- D) Maintenance of homeostasis

2. In an experiment testing how fertilizer affects tomato plant growth, the amount of fertilizer applied is the:

- A) Dependent variable
- B) Independent variable
- C) Control group
- D) Conclusion

3. Identify the correct biological hierarchy from smallest to largest:

- A) Organelle → Cell → Tissue → Organ → Organ System
- B) Tissue → Cell → Organelle → Organ → Organism
- C) Organ → Tissue → Cell → Organelle → Molecule
- D) Molecule → Tissue → Cell → Organ → Organ System

4. Which of the following best describes an ecosystem?

- A) All the individuals of a single species living in an area
- B) A community of organisms interacting with their physical environment
- C) The entire planet Earth and its atmosphere
- D) A group of cells working together for a common function

5. Evolution by natural selection occurs when:

- A) Individuals develop new traits during their lifetime
- B) Populations adapt as individuals with advantageous traits survive and reproduce
- C) All organisms stop changing to become perfect
- D) Random mutations create new species instantly

Module 02: Chemistry of Life

1. An element with 6 protons, 6 neutrons, and 6 electrons has an atomic number of:

- A) 18
- B) 12
- C) 6

- D) 0

2. Covalent bonds are formed when atoms:

- A) Share one or more pairs of electrons
- B) Transfer electrons completely
- C) Are held together by magnetic attraction
- D) Lose protons to become ions

3. Which of the following is NOT a property of water?

- A) It is a universal solvent for polar substances
- B) It expands when it freezes, becoming less dense
- C) It strongly repels other water molecules (low cohesion)
- D) It resists rapid temperature changes

4. A substance with a pH of 2 would be classified as:

- A) Weakly basic
- B) Neutral
- C) Strongly acidic
- D) Strongly basic

5. Carbon is the fundamental element of organic molecules because it:

- A) Forms only ionic bonds
- B) Can form up to four stable covalent bonds
- C) Is the most abundant element in the universe
- D) Dissolves easily in nonpolar solvents

Module 03: Biomolecules

1. The process that joins monomers into polymers by removing water is called:

- A) Hydrolysis
- B) Oxidation
- C) Dehydration synthesis

- D) Fermentation
2. Which biomolecule is the primary source of quick cellular energy?
- A) Lipids
 - B) Proteins
 - C) Nucleic acids
 - D) Carbohydrates
3. Amino acids are linked together in proteins by which type of bond?
- A) Peptide bond
 - B) Hydrogen bond
 - C) Glycosidic linkage
 - D) Ester bond
4. Saturated fats are chemically distinct from unsaturated fats because they:
- A) Contain one or more double bonds
 - B) Are liquid at room temperature
 - C) Contain the maximum number of hydrogen atoms
 - D) Are found primarily in plants
5. The loss of a protein's functional 3D shape due to heat or pH changes is known as:
- A) Translation
 - B) Denaturation
 - C) Replication
 - D) Hydrolysis

Module 04: Cellular Function

1. Which statement represents a core principle of cell theory?
- A) All cells have a nucleus
 - B) All living organisms are composed of cells
 - C) Cells differ in their basic chemical composition

- D) Spontaneous generation creates new cells
2. Eukaryotic cells differ from prokaryotic cells because eukaryotes possess:
- A) A plasma membrane
 - B) DNA
 - C) Ribosomes
 - D) A membrane-bound nucleus
3. Which organelle is responsible for sorting, modifying, and packaging proteins?
- A) Smooth endoplasmic reticulum
 - B) Golgi apparatus
 - C) Lysosome
 - D) Mitochondrion
4. Evidence for the endosymbiotic theory includes the fact that mitochondria:
- A) Have their own circular DNA and ribosomes
 - B) Lack membranes
 - C) Are larger than the nucleus
 - D) Are found only in plant cells
5. Which structure provides rigidity to plant cells but is absent in animal cells?
- A) Cell membrane
 - B) Mitochondria
 - C) Cell wall
 - D) Ribosome

Module 05: Membranes

1. The fluid mosaic model describes the cell membrane as:
- A) A solid layer of protein
 - B) A flexible bilayer of phospholipids with embedded proteins
 - C) A rigid structure made of cellulose

- D) A single layer of lipids
2. Facilitated diffusion differs from simple diffusion in that it:
- A) Requires cellular energy (ATP)
 - B) Moves molecules against a concentration gradient
 - C) Requires transport proteins
 - D) Occurs only in plant cells
3. If a red blood cell is placed in a hypotonic solution, water will:
- A) Move into the cell, causing it to swell
 - B) Move out of the cell, causing it to shrink
 - C) Move in and out at equal rates
 - D) Stop moving completely
4. The sodium-potassium pump is an example of active transport because it:
- A) Uses ATP to move ions against their concentration gradient
 - B) Moves ions down their concentration gradient without energy
 - C) Requires no energy input
 - D) Moves only water molecules
5. Glycoproteins on the cell surface function primarily in:
- A) Energy storage
 - B) Cell-to-cell recognition
 - C) Structural support
 - D) DNA replication

Module 06: Metabolism

1. The primary energy currency used by cells to power work is:
- A) Glucose
 - B) ATP
 - C) DNA

- D) NADH

2. Enzymes catalyze reactions by:

- A) Increasing the temperature of the reactants
- B) Lowering the activation energy required
- C) Changing the free energy of the reaction
- D) Being consumed in the reaction

3. The overall purpose of cellular respiration is to:

- A) Create glucose from sunlight and water
- B) Break down glucose to produce ATP
- C) Use oxygen to build new proteins
- D) Produce oxygen as a waste product

4. Oxygen's role in aerobic respiration is to:

- A) Donate electrons to the transport chain
- B) Serve as the initial fuel source
- C) Act as the final electron acceptor
- D) Catalyze the breakdown of glucose

5. The metabolic process that builds larger molecules from smaller ones (consuming energy) is called:

- A) Catabolism
 - B) Hydrolysis
 - C) Anabolism
 - D) Digestion
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Part B: Fill in the Blank (11 points)

Complete each statement with the correct terminology. Each question is worth 1 point.

1. The tendency of an organism to maintain a stable internal environment is called _____.
 2. A chemical bond formed by the transfer of electrons from one atom to another is a(n) _____ bond.
 3. The chemical reaction that breaks down polymers into monomers by adding water is known as _____.
 4. The four levels of protein structure are primary, secondary, _____, and quaternary.
 5. The _____ endoplasmic reticulum is studded with ribosomes and is involved in protein synthesis.
 6. Membrane-bound sacs containing digestive enzymes for breaking down waste are called _____.
 7. The passive movement of water across a semipermeable membrane is called _____.
 8. A solution with the same solute concentration as the cell is described as _____.
 9. The concept that an enzyme changes shape to fit its substrate is called the _____ model.
 10. The anaerobic process by which cells produce ATP without oxygen is called _____.
 11. The large organelle that contains the cell's genetic material (DNA) and controls cellular activities is the _____.
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Part C: Free Response (9 points)

Choose THREE of the following five questions. Answer in the provided spaces below. Each response is worth 3 points.

1. **Water and Life.** Identify three properties of water that are crucial for life. Provide a brief biological example for each.
2. **The Protein Path.** Describe step-by-step how a protein is made and secreted from a cell, naming the organelles involved.
3. **Moving Across Membranes.** Explain the difference between passive and active transport. Give one example of each.
4. **Respiration vs. Fermentation.** Compare aerobic respiration and fermentation. Which requires oxygen? Which produces more ATP? Where do they happen in the cell?
5. **Macromolecule Structure.** Choose two macromolecules (carbohydrates, lipids, proteins, nucleic acids). For each, name its building block (monomer) and one major function.

Response 1

Question Number: _____

Response 2

Question Number: _____

Response 3

Question Number: _____

Feedback (Optional)

A. Is there anything else you would like to share (e.g., questions, comments, or notes for the professor)?

B. What percentage grade do you estimate you earned on this exam? How confident are you in that estimate (1–10)?

End of Exam 01